

Mehul Sarin

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EDUCATION

McMaster University

Expected May 2029

Bachelor of Science, Mechanical Engineering | CGPA: 3.8

Hamilton, ON

Honours: Dean's List; McMaster Scholar Award (\$10,000)

Extracurricular: Academy Badminton (Doubles) & Intramural Badminton/Football | Private Peer Tutor | Residence Tour Guide | Barbell Club | 2x Case Competition Finalist

Relevant Coursework: Engineering Design & Prototyping, Engineering Mathematics (Calc I–IV, Vector Calculus, Differential Equations, Linear Algebra), Mechanics, Waves & Electromagnetism, Engineering Chemistry

EXPERIENCE & LEADERSHIP

Omazzi Print

May 2026 – Aug 2026

Manufacturing Engineering Intern

Toronto, ON

- Reduced production-line downtime ~**12%** through root-cause analysis and corrective action on recurring equipment failures across 3 production lines.
- Cut print cycle time **32%** (5.0s → 3.4s per page) by analyzing workflow data and optimizing equipment setup to reduce process variation.
- Designed and fabricated a custom transfer fixture in **SolidWorks** to automate component handoff from the primary press to finishing machines, cutting manual handling time and changeover defects.

McMaster Concrete Canoe Team

Oct 2025 – Present

Concrete Canoe Construction Member

Hamilton, ON

- Analyzed structural loads using SolidWorks Simulation (FEA) to validate wall thickness and reinforcement layout for a 146.5 kg, 5.6 m canoe, hitting a **13.9 mm** wall-thickness target under competition load limits.
- Optimized aggregate ratios during concrete mix design and casting to balance **compressive strength** against the canoe's weight target while maintaining quality control.
- Delivered a **7th-place** overall finish among 500+ competing students nationally, contributing to a 1st-place Enhanced Focus Area submission and the Most Improved Team Award two years running.

Fiverr & Upwork

Sept 2021 – Present

Freelance 3D CAD Designer

Remote

- Designed **20+** mechanical parts, assemblies, and technical drawings in **Autodesk AutoCAD** (brackets, enclosures, functional assemblies) for 15+ clients, converting sketches and requirements into manufacturing-ready CAD models.
- Reduced client revision cycles ~**25%** and held 95%+ on-time delivery through a structured design-review process, growing the practice to \$300+ in revenue alongside full-time coursework.

PROJECTS

3D-Printed Prosthetic Exo-Arm | Design Competition Finalist

2026

- Designed and fabricated a **77 g** modular prosthetic exo-arm using SolidWorks CAD and 3D printing for users with arthritis and limited hand dexterity, staying well under the 500 g design limit.
- Delivered a working, low-cost device built for under **\$10** in PLA, validated through prototyping and end-user testing, successfully enabling grip-free interaction with everyday objects.

Robotic End-Effector for Warehouse Automation | Team Manager

2025

- Led a 5-person team to design and fabricate, using **Autodesk Inventor** and Python, a 3D-printed robotic gripper for a Q-Arm, achieving **5-of-6** success across test objects varying in shape, weight, and center of mass, modeling real warehouse pick-and-place conditions.
- Improved grip reliability by redesigning the claw to a **45° front-grasp** configuration and adding high-friction foam after identifying slippage and a servo-gear fault through iterative testing.
- Built modular **Python** control software for automated pick-and-place (barcode ID, order processing, CSV logging), debugging conflicting motion functions to improve positioning accuracy.

Hospital Wastewater Filtration System | Materials Engineering Research Project

2026

- Developed a low-cost filtration fiber to capture **5–10 µm** particles, addressing hospital wastewater contamination in Byron Bay, Australia, where the antibiotic ciprofloxacin was entering local waterways.
- Selected Low-Density Rigid Polymer Foam as the optimal filtration material (ranked **15.5/20**) via Ashby analysis in Granta EduPack across strength, stiffness, density, cost, and durability.
- Optimized a **7 mm × 15 cm** filtration element (target porosity 0.470, pore size 0.008 mm) by balancing filtration performance against strength requirements, validated through life-cycle and Eco-Audit analysis.

SKILLS

CAD & Design: SolidWorks (CSWA Certified), Autodesk Inventor, Autodesk AutoCAD, GD&T, Detail & Assembly Drawings, PLM/PDM

Simulation & Analysis: ANSYS (FEA & CFD), ANSYS Fluent, ANSYS Mechanical, SolidWorks Simulation, Simulink

Programming: MATLAB, Python, C++, C, VBA, HTML, JavaScript, Microsoft Office 365 (Excel, VBA), Google Suite

Manufacturing: CNC & Vertical Milling, Drill Press, Power Saw, 3D Printing (FDM), Soldering, Hand Tools

Core Competencies: Mechanical Design, Structural Analysis (FEA), Material Selection, Root Cause Analysis, Process Improvement